Article

Dendrite initiation and propagation in lithium metal solid-state batteries

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All-solid-state batteries with a Li anode and ceramic electrolyte have the potential to deliver a step change in performance compared with today's Li-ion batteries[1,2]. However, Li dendrites (filaments) form on charging at practical rates and penetrate the ceramic electrolyte, leading to short circuit and cell failure[3,4]. Previous models of dendrite penetration have generally focused on a single process for dendrite initiation and propagation, with Li driving the crack at its tip[5–9]. Here we show that initiation and propagation are separate processes. Initiation arises from Li deposition into subsurface pores, by means of microcracks that connect the pores to the surface. Once filled, further charging builds pressure in the pores owing to the slow extrusion of Li (viscoplastic flow) back to the surface, leading to cracking. By contrast, dendrite propagation occurs by wedge opening, with Li driving the dry crack from the rear, not the tip. Whereas initiation is determined by the local (microscopic) fracture strength at the grain boundaries, the pore size, pore population density and current density, propagation depends on the (macroscopic) fracture toughness of the ceramic, the length of the Li dendrite (filament) that partially occupies the dry crack, current density, stack pressure and the charge capacity accessed during each cycle. Lower stack pressures suppress propagation, markedly extending the number of cycles before short circuit in cells in which dendrites have initiated.

The ceramic electrolyte microstructure obtained from operando X-ray computed tomography (XCT) and focused ion beam scanning electron microscopy (FIB-SEM), in combination with the local (grain boundary) fracture strength from microcantilever measurements, is used to predict the critical current for initiation. Stack pressures of a few MPa do not affect the critical initiation current, but values above 50 MPa, typical in the literature[10–12], lower it. The electric field and current density enhancement previously suggested to occur in subsurface pores[13] is reduced when pores are close together, which also affects critical current predictions.

In practical cells, pressures at the Li anode may be applied externally or develop internally. By varying stack pressure, we can investigate systematically the effect of pressurizing Li anodes during charge on crack propagation. Although Li filaments elongate within a dry crack during plating and shrink during stripping, stack pressure induces net filament elongation on each cycle. We identify a critical length of lithium filling within a dry crack that causes the crack tip to propagate. Moderate stack pressures (7 MPa) cause Li filaments to reach this critical propagation length after a few cycles at moderate current density. By lowering stack pressure, cycle life before short circuit can be extended substantially. A Li_6PS_5Cl solid electrolyte cell with a Li anode, cycled at a plating current density of 4.0 mA cm^{-2} and stripped at 0.05 mA cm^{-2} (for a capacity of 0.5 mAh cm^{-2} per half-cycle), failed after 35 cycles under 7 MPa of stack pressure, whereas an identical cell under atmospheric pressure (0.1 MPa) had its cycling prolonged five-fold, avoiding short circuit for 170 cycles. The results demonstrate that dendrites can be inhibited by suppressing their propagation. Inhibiting dendrite penetration can be based on suppressing either initiation or propagation, the former by increasing the local fracture strength, as well as minimizing pore size and controlling pore proximity, and the latter by maximizing electrolyte fracture toughness and minimizing pressure on the Li anode.

Operando XCT

We used continuous operando XCT scanning with vastly improved time resolution compared with previous studies[14,15] to follow the initiation and propagation of cracks during constant-current plating. XCT data are reported on an argyrodite sample sintered under identical conditions as the material used to determine mechanical properties, ensuring consistency of the parameters used for modelling. Lithium plating first creates a spallation at the edge of the metal electrode, followed by the

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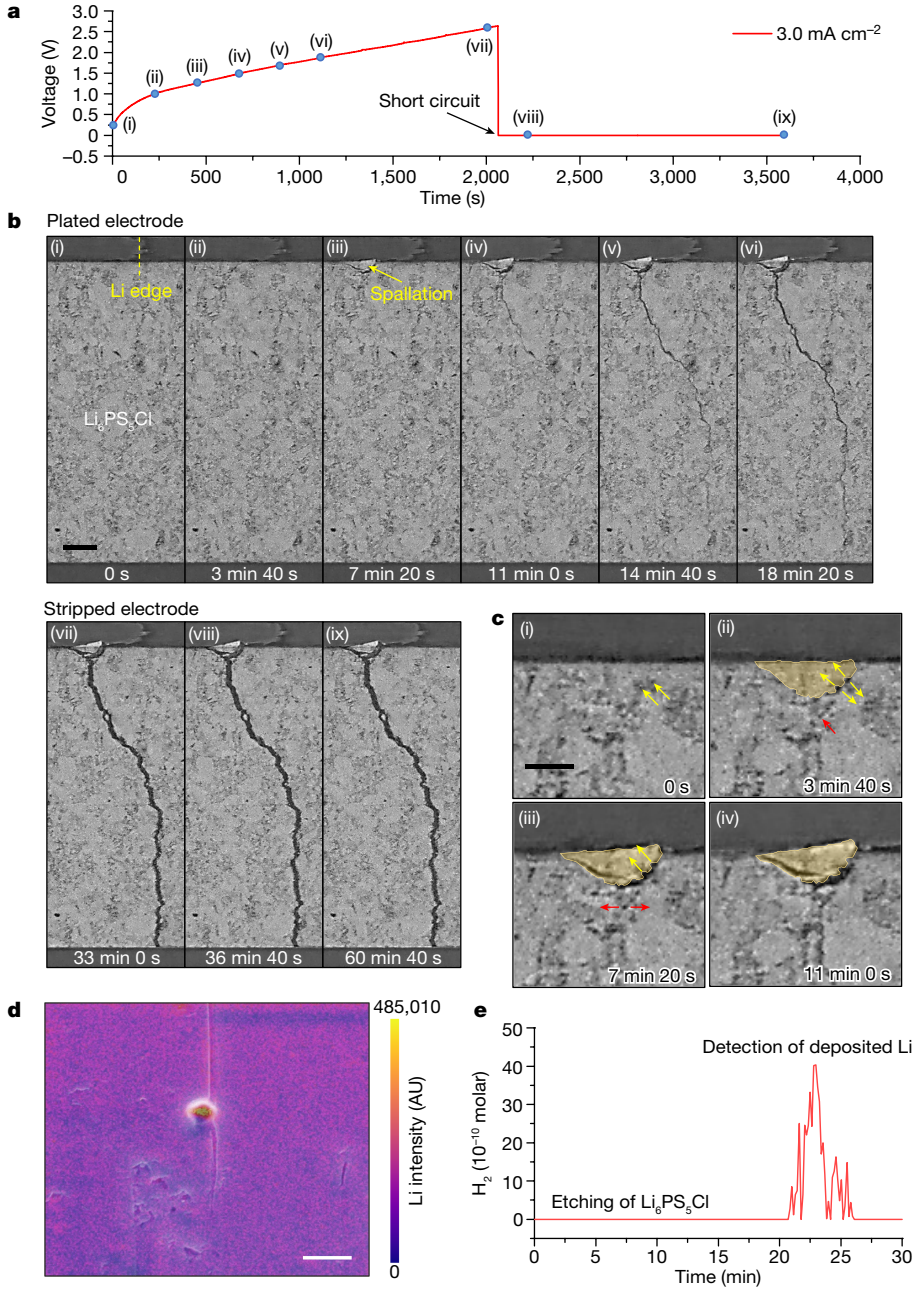


Fig. 1 | Operando XCT virtual cross-sections during plating of a Li/Li_6PS_5Cl/Li cell showing the development of a dendrite crack from initiation through propagation to complete short circuit. Combined FIB-SEM with SIMS as well as mass spectrometry provides evidence supporting the presence of Li in subsurface regions of the solid electrolyte. a, Voltage versus time for a Li/Li_6PS_5Cl/Li cell, plated at a current density of 3.0 mA cm^{-2} to a capacity of 3.0 mAh cm^{-2} under 7 MPa stack pressure. The points indicate times at which XCT scans, shown in **b** (i)–(ix), terminated. **b**, Virtual cross-sectional image slices show the growth of a crack through an initially pristine electrolyte. A dendrite initiates at the lithium-plating (top) surface (i)–(iii), forming a transverse crack that propagates until the cell short circuits (iv)–(ix). See also Supplementary Video 1. Scale bar, 100 µm. **c**, Magnified images of the region near the plated electrode in **b**, pristine (i) and after successive plating (ii)–(iv), showing formation of the spallation and transverse crack. Yellow arrows indicate pores associated with forming the spallation and red arrows indicate pores associated with propagation of the transverse crack. The spalled-out solid electrolyte is shaded in yellow to illustrate its movement. Scale bar, 50 µm. **d**, FIB-SEM image of a subsurface pore within a Li_6PS_5Cl electrolyte disc after plating. The corresponding Li^+ SIMS mapping is overlaid, demonstrating the presence of Li metal in the subsurface pore. Scale bar, 2 µm. AU, arbitrary units. **e**, Detection of H_2 in the gas phase from a Li_6PS_5Cl solid electrolyte disc etched with excess LiOH solution after Li plating and removal of the electrodes; the delayed response indicates Li deposition primarily in the subsurface region of the electrolyte (see also Supplementary Fig. 9).

formation of a transverse crack that propagates across the electrolyte to the other electrode (Fig. 1b). This process accords with our earlier studies[3] in that the crack reaches the other electrode before the Li dendrite, indicated by the lack of short circuit until after step (vii) (Fig. 1a). Li also plates on the working electrode, which thickens as a result, not just in the crack. To demonstrate that Li in the transverse crack reaching the counter electrode is responsible for the short circuit and not any other Li filaments elsewhere, we show XCT results in Supplementary Fig. 8, in which we follow the progression of Li along the crack. The short circuit occurs at the point at which the Li reaches the counter electrode.

Images (i)–(iv) in Fig. 1c show the earliest morphological changes. By 7 min 20 s, a piece of argyrodite has been spalled out, detaching from the bulk electrolyte, (iii). The image before detachment, (ii), shows that the fracture pathway of the spallation connects pre-existing pores 20–50 μm from the interface (two are indicated by yellow arrows). Comparing (i) and (ii) shows that the pre-existing pores enlarge as plating proceeds, suggesting that they are being filled with Li. Pre-existing pores are also observed along the fracture pathway of the transverse crack, marked by red arrows in (iii).

The deposition of Li in subsurface pores was further demonstrated by two techniques. A FIB-SEM cross-sectional image showing the pores was combined with secondary-ion mass spectrometry (SIMS) analysis to identify Li (Fig. 1d). The results show a subsurface pore filled with Li metal within the Li_6PS_5Cl electrolyte after plating. Second, subsurface Li metal was detected in a Li_6PS_5Cl disc after plating, removal from the cell and etching with LiOH solution while concurrently performing mass spectrometry (Fig. 1e), as detailed in Methods. The lag in H_2 detection is consistent with Li being primarily deposited in subsurface pores.

Overall, the results in Fig. 1 suggest two stages of dendritic failure, crack initiation and crack propagation.

Crack initiation based on pore filling

The model (Fig. 2) simulates a subsurface pore, as a spherical cavity, connected to the exterior of the electrolyte by a pre-existing microcrack, modelled as a cylindrical void space normal to the electrode surface. On plating, Li deposits first at the top surface of the microcrack, progressively filling it and the pore (Fig. 2a). It has been suggested that Li nucleation can occur in subsurface pores because of some electronic conductivity[13,16,17]. Whatever the process of pre-filling, the entire pore-crack assembly fills early on initial plating, leading to a lithium-filled configuration shown in Fig. 2b. Further Li deposition occurs across the entire Li–electrolyte interface (pore and microcrack surface). Because the defect is already occupied, this deposition induces a strain within the defect structure, which is accompanied by a pressure buildup. The net effect is Li extrusion back along the microcrack, to accommodate the freshly deposited Li. Because lithium metal is a viscoplastic solid, its motion along the narrow microcrack is analogous to a non-Newtonian pipe flow and is largely controlled by the current density in the subsurface pore. At sufficiently high plating rates, the high pressure drop associated with this viscoplastic flow is capable of inducing electrolyte fracture. Fracture near subsurface pores associated with lithium flow through microcracks thus underpins the initiation process.

Estimation of the critical current density for dendrite crack initiation, CCD_{init}, necessitates viscoplastic property data for Li metal, as well as a fracture criterion for the solid electrolyte. Recent experiments quantifying the power-law creep of Li have been incorporated into the model, as detailed in the Supplementary Information[18,19]. The fracture criterion requires information about how the ceramic responds to microscopically applied stress[20]. Such local fracture characteristics cannot be quantified accurately by standard mechanical tests, which sample macroscopic properties averaged across several grains[21]. Instead, the local fracture strength of grain boundaries in the argyrodite was measured by microcantilever bending, as depicted in Fig. 2c (see also Methods and Supplementary Fig. 19). The smoothness of the microcantilever fracture surfaces suggests intergranular fracture (see Supplementary Fig. 20), although this is indirect evidence and not conclusive. The situation may be different for oxide solid electrolytes, which have different mechanical properties.

The model for crack initiation and accompanying calculations are detailed in the Supplementary Information. The results are presented graphically in Fig. 2d,e. Using the pore dimensions and proximity to the surface from XCT (Fig. 1), as well as FIB-SEM images (Supplementary Fig. 10), calculations showing how hydrodynamic pressure in the pore depends on the plating current density are presented in Fig. 2d. CCD_{init} is determined by the local fracture strength, shown by the dashed line in Fig. 2d. At a sufficiently high current density, the pore pressure exceeds the local fracture strength, inducing fracture.

The current density at the pores is a balance of two factors. The field enhancement at the tips of a conductor effect[9,22] acts to increase the current density, whereas the presence of neighbouring pores spreads the current, both of which are important in relating the overall applied current density to that at the pores (see Supplementary Fig. 2).

The turquoise line in Fig. 2d represents a smooth-tipped microcrack without a larger terminal pore. The other lines in Fig. 2d show how the presence of pores and their increasing size induces a higher extrusion rate of Li at a given applied current density, and hence higher pressure, explaining why subsurface pores are the weaknesses most likely to initiate cracks. Figure 2e shows that decreasing pore size or increasing the local fracture strength increases CCD_{init}.

On the basis of the size, population density and surface proximity of pores and the local fracture strength determined for Li_6PS_5Cl with XCT, SEM and microcantilever measurements (see Supplementary Information), the model predicts that CCD_{init} is approximately 1.0 mA cm^{-2}, in line with typical experimental values[4,23].

Crack propagation based on wedge opening

The observations of dry cracks that lengthen and widen on plating, in which Li fills the base but is absent from the tip, suggests a wedge-opening propagation mechanism.

The images in Fig. 1b and Supplementary Fig. 12 indicate that dendrite propagation across the electrolyte occurs within an approximately planar crack, represented schematically in Fig. 3a. Lithium plating occurs along the crack interior that contacts the crack-filling metal. Li deposition at the lateral interfaces (blue surfaces in Fig. 3b) induces a pressure buildup and a resulting viscoplastic flow that drives motion of the dendrite tip into the void space and Li extrusion back into the bulk Li anode. Applied stack pressure enhances Li flow towards the crack tip and inhibits extrusion. Extrusion only occurs when the pressure induced by the plating current—which is generally non-uniform along the length of the dendrite—exceeds the stack pressure.

By modelling the stress field near the dry crack tip, the strain energy release rate can be calculated, that is, the energy available to lengthen the crack, quantified by the J-integral[24]. The criterion for fracture, and therefore crack propagation, is determined by comparing the calculated J-integral to the critical strain energy release rate required for propagation, a material property dependent on the electrolyte's fracture toughness, Young's modulus and Poisson's ratio. The fracture toughness and Young's modulus of the Li_6PS_5Cl solid electrolyte were measured by nanoindentation, as described in Methods. The propagation model is detailed in the Supplementary Information.

The operando XCT micrograph in Fig. 1b (iv) shows a crack in the process of propagation, which has initiated but not yet reached the counter electrode. From its approximate dimensions, 200 μm long and 2 μm wide, the stress field at the crack tip was modelled quasi-statically during the continuous growth of the Li dendrite at a cell-level current density of 3.0 mA cm^{-2} (typical of a practical charging rate and commensurate with the current used to collect the XCT in Fig. 1); see Supplementary Information for details. As the Li dendrite grows in the crack, the strain energy release rate (J-integral) rises super-linearly with the Li dendrite length. Increasing area of Li–electrolyte contact within a dendrite increases both the flux of Li into the dendrite and the area of the electrolyte under stress from pressurized crack-filling Li (Fig. 3c). Because pressures within the dendrite depend on the Li flow, the magnitude of the J-integral for a given dendrite length is greater for higher current densities, as shown for a fixed dendrite length of 115 μm in Fig. 3d.

The critical J-integral for crack propagation in Li_6PS_5Cl is shown by the dashed lines in Fig. 3c,d. Figure 3c shows the effect of dendrite length:

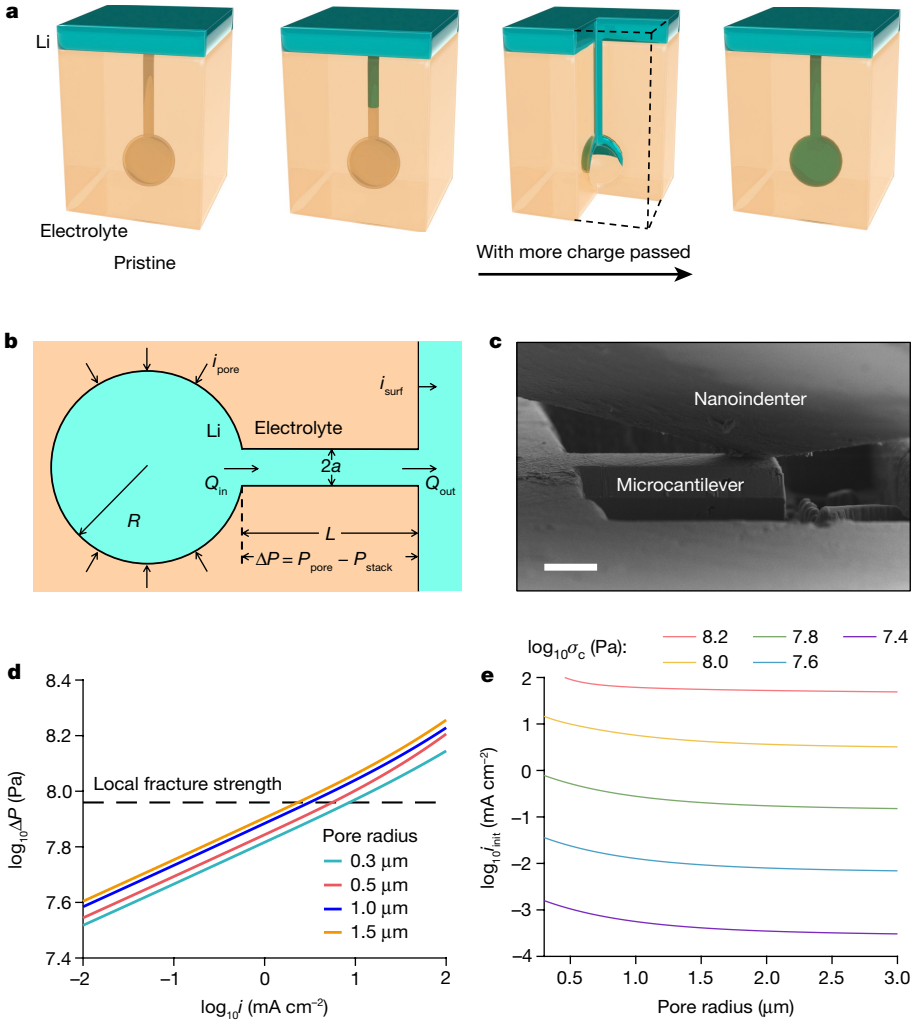


Fig. 2 | Schematics and implications of the dendritic crack initiation process. Higher current density leads to greater pressure in the pore, with cracking occurring when the pressure exceeds the local (grain boundary) fracture strength of the solid electrolyte. a, Schematic of a pore connected to the lithium–solid electrolyte interface through a microcrack and being filled by Li deposition. **b**, Schematic of the filled subsurface pore-microcrack system, labelled with parameters relevant to the dendrite initiation model: P_{stack} is the stack pressure on the cell. During constant-current charging, Li is plated with a current density of i_{surf}, whereas the pore interior experiences a different current density i_{pore}. Because the lithium is incompressible and ductile, i_{pore} induces extrusion of lithium towards the surface. As a consequence of this current-induced lithium extrusion, a pressure, P_{pore}, builds up within the pore in excess of the stack pressure, P_{stack}. The pore population density was used to calculate i_{pore} and i_{surf} (see Supplementary Figs. 2 and 11) **c**, SEM image depicting the measurement of local fracture strength by microcantilever bend tests (see also Methods and Supplementary Fig. 19). Scale bar, 10 µm. **d**, The dependence of hydrodynamic pressure in the pore on current density, for various pore radii: 1.5 µm (orange), 1.0 µm (blue), 0.5 µm (red) and with no pore, that is, a microcrack that terminates with a hemispherical tip, $R = a$, in which a is 300 nm (turquoise). For simulations, the microcrack length L was 5 µm and the separation distance between pores was 10 µm (see Supplementary Fig. 11). The dashed line shows the local fracture strength, σ_c, of the electrolyte determined by the microcantilever experiments. **e**, Plot showing how the critical current density for crack initiation varies with pore size for different local fracture strengths.

for a plating current density of 3.0 mA cm^{-2} under 7 MPa stack pressure, the Li only needs to fill half the crack length (approximately 115 µm) before the dry crack lengthens, consistent with the wedge-opening propagation seen in Fig. 1 and our previous work[3]. When the stack pressure is reduced, dendrites must be longer to propagate the dry crack. For a dendrite length of 115 µm and stack pressure of 7 MPa, crack propagation occurs at 2.4 mA cm^{-2}, whereas removal of the stack pressure increases this substantially, to 4.2 mA cm^{-2} (Fig. 3d). Stack pressure has a great influence on how lithium deposition drives the growth of dry cracks.

Crack propagation can be inhibited by toughening the electrolyte[21,25] (Fig. 3e), because increasing fracture toughness increases the J-integral required for crack extension. With a higher fracture toughness, crack propagation requires higher current densities or longer Li dendrites (Fig. 3e).

When does propagation lead to a short circuit?

The capacity plated and stripped per half-cycle and the stack pressure affect whether crack propagation will result in a short circuit. Taking the 200-µm-long, 2-µm-wide crack from the XCT section shown in Fig. 1b (iv), simulations to determine the net Li dendrite growth after a single cycle as a function of capacity were performed for applied stack pressures of 7 MPa, 1 MPa and 0 MPa (Fig. 4a). Simulated dendrite length during repeated cycling is shown in Fig. 4b. The net dendrite-length change during a plating/stripping cycle over a given capacity varies with the applied stack pressure. Stack pressure promotes Li flow towards the counter electrode and inhibits extrusion, resulting in greater net growth. Cycling at a stack pressure of 7 MPa (orange) results in a substantial cumulative length change that soon drives crack propagation (dashed line in Fig. 4b), whereas lower stack pressures reduce the net

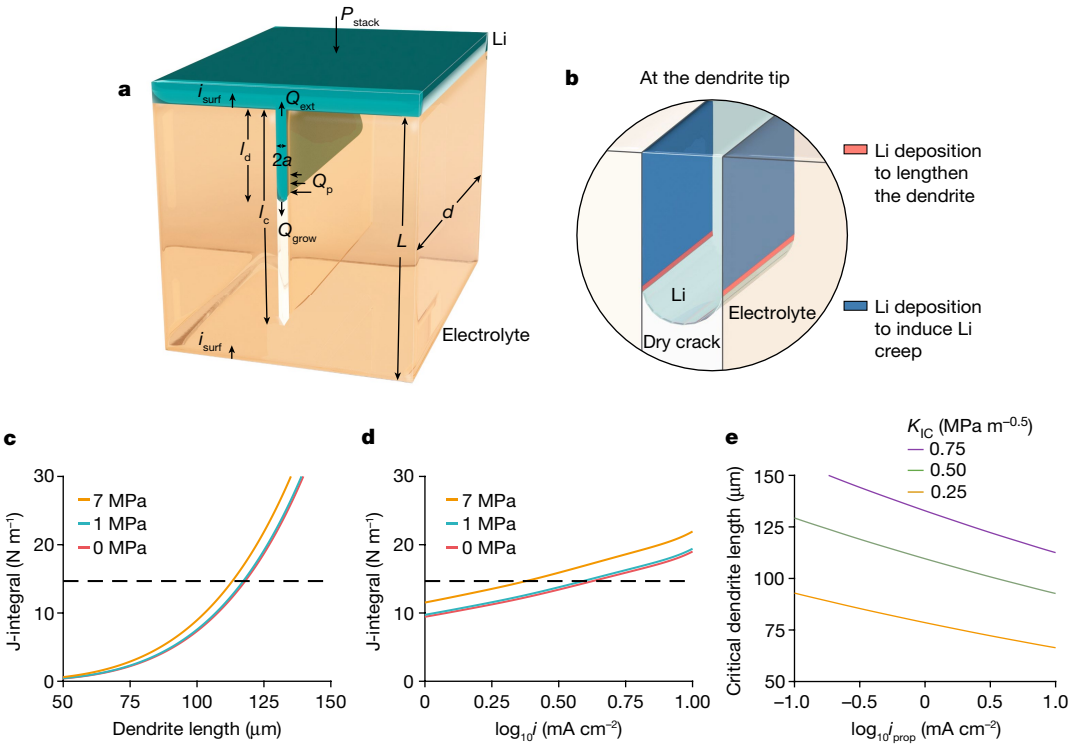


Fig. 3 | Dendrite crack propagation. Longer Li dendrites, higher currents and greater stack pressures promote crack propagation. Li dendrites do not need to fill a crack completely to meet the fracture criterion required for crack lengthening. a, Schematic of a planar dendritic crack: length l_c and width $2a$, partially filled by a Li dendrite of length l_d. Cell stack pressure of P_{stack}, cell-level current density of i_{surf}. Li plating takes place at all Li–electrolyte interfaces. **b**, Li deposition in which the crack surface in contact with the metal (blue region) drives viscoplastic flow that causes an extrusion flux Q_{ext} back into the bulk Li at the base of the dendrite and a flow Q_{grow} that extends the Li dendrite further within the dry crack. Deposition at the three-phase contact line at which Li, electrolyte and free space meet (red region) contributes directly to lengthening of the Li dendrite. **c**, J-integral at the crack tip as a function of dendrite length. **d**, J-integral at the crack tip as a function of the current density for Li plating (dendrite length l_d = 115 μm). For a given dendrite length and stack pressure, the intersection with the dashed line defines a current density, i_{prop}, that will propagate the dendrite crack. **e**, The effect of fracture toughness, K_{IC}, on applied current density, i_{prop}, and lithium dendrite length required to lengthen a dry crack (shown for a 200-μm-long dry crack and zero stack pressure).

propagation markedly (Fig. 4b). For a given stack pressure and current density, the number of cycles to short circuit depends on the capacity per half-cycle and the electrolyte thickness. This effect of stack pressure on the net accumulation of Li in the electrolyte while cycling was further demonstrated experimentally by quantifying the amount of Li remaining within the solid electrolyte after cycling at different stack pressures using mass spectrometry (see Supplementary Fig. 13). Higher pressures and cycle numbers lead to more residual Li in the solid electrolyte after stripping.

As stack pressures tend to zero, very little net propagation occurs (Fig. 4a). The dendrite length oscillates during plating and stripping, indicating that the cell can be cycled more extensively before failure. To demonstrate this, three-electrode Li/Li_6PS_5Cl cells were assembled, as described in Methods, and plated at 4.0 mA cm^{-2} under stack pressures of 0.1 MPa (ambient) and 7 MPa (Fig. 4c,d). The cells were cycled until short circuit. A drop in voltage at the end of plating was observed in the first few cycles at both stack pressures, consistent with previous work indicating an increase in contact area as Li penetrates the electrolyte. The cell under ambient pressure achieved a fivefold increase in the number of cycles, corresponding to 170 cycles (Fig. 4c), compared with the cell under 7 MPa stack pressure, which short-circuited during the 35th cycle. The effect of pressure on short circuit is further verified at higher pressures and by quantifying the amount of capacity passed (Li deposited) to short circuit on a single plating; see Supplementary Figs. 14 and 16, respectively. In both figures, higher pressures lead to earlier short circuit. Kinzer et al.[26] demonstrated critical currents of several hundreds of mA cm^{-2} when using a molten Li anode owing to the vastly improved viscoplasticity of the metal anode, as well as a lack of applied stack pressure. Similarly, low temperatures increase the resistance of Li to flow and lead to early cell failure, as shown in Supplementary Fig. 17, in which the same experiment as in Fig. 4d was carried out but at 0 °C, with the cell failing after only 17 cycles. Of course, current density as well as capacity play a role in the pressure dependence of short circuits, as seen in recent studies of stack pressure effects on Li dendrites in solid electrolytes[27–29]. For example, if sufficiently low current densities are used, it is possible to sustain cycling under pressures of several MPa (refs. 4,29).

To focus on the fundamental understanding of dendrites, all cells contained Li metal electrodes in the pristine state. Anodeless conditions and cycling efficiencies would require a separate study.

Implications

Li dendrites and short circuits can be suppressed by inhibiting either the initiation or the propagation of cracks in the solid electrolyte, and the means of doing so differ. For a sintered Li_6PS_5Cl argyrodite electrolyte, using measurements of the microscopic (grain boundary) fracture strength, pore size and population density, a critical current for initiation of approximately 1.0 mA cm^{-2} was calculated. The experimentally reported range is 0.5–3.0 mA cm^{-2} (refs. 4,23). Previous efforts have often sought to increase the critical current by raising the macroscopic fracture toughness of the electrolyte to prevent dendrite initiation. Future strategies should consider local fracture strength. Also, lowering pore size and minimizing the number of isolated subsurface

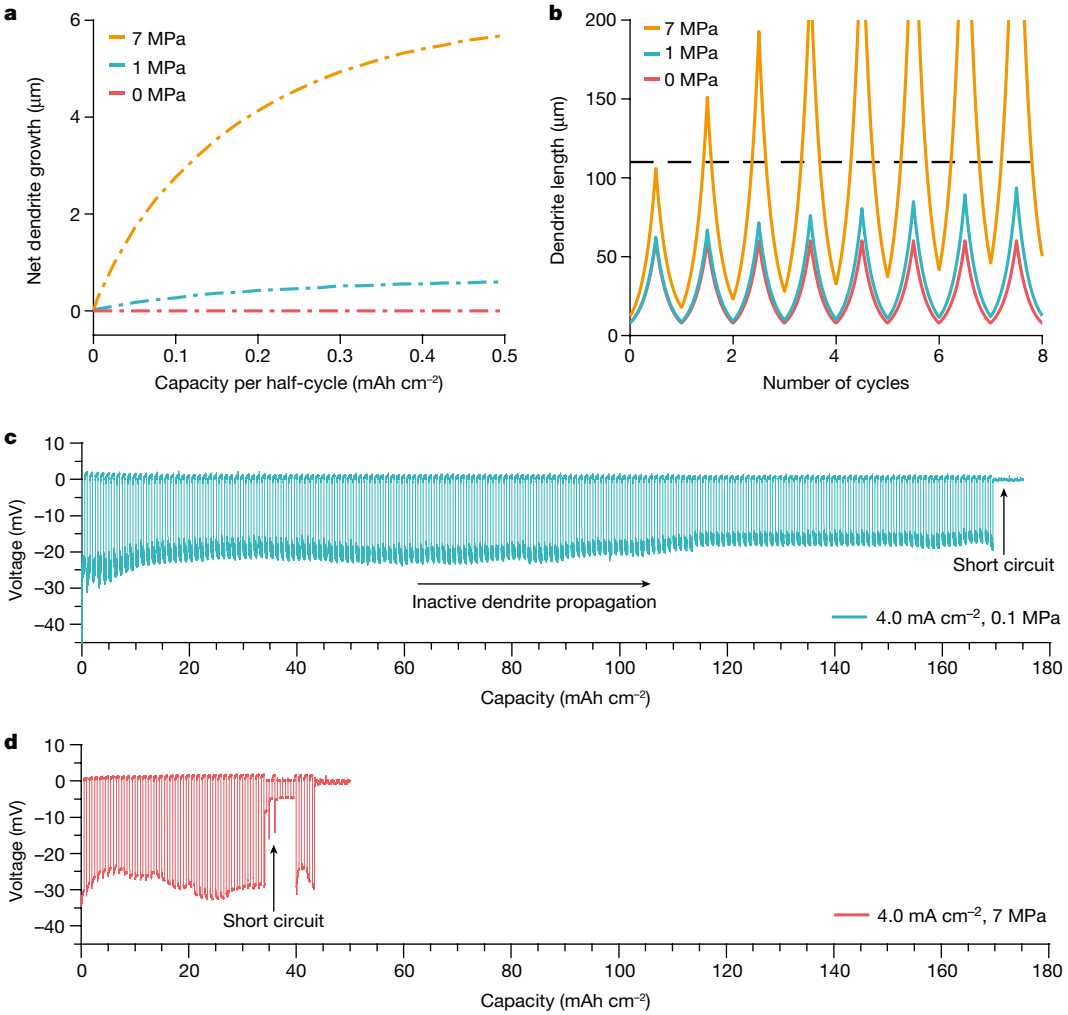


Fig. 4 | Propagation of a Li dendrite under various stack pressures. Larger stack pressures reduce the number of cycles to short circuit. Removal of applied stack pressure can enable prolonged cycling. a, The net change in Li dendrite length within the crack on one charge–discharge cycle as a function of capacity under various stack pressures. **b**, Cumulative dendrite length during cell cycling under various stack pressures. As stack pressures reduce, the net lengthening of the Li dendrite per cycle tends to zero, inhibiting both crack propagation and short circuiting by the dendrite. **c**,**d**, Cycling of a three-electrode Li/Li_6PS_5Cl cell with current densities of 4.0 mA cm^{-2} for plating, 0.05 mA cm^{-2} for stripping (to avoid void formation), both for a capacity of 0.5 mAh cm^{-2}, under ambient pressure (0.1 MPa) (**c**) and 7 MPa of pressure (**d**).

pores will increase the critical current for dendrite initiation. Even if dendritic crack initiation cannot be avoided, it is possible to prevent short circuit for many cycles, by suppressing crack propagation or by limiting the net growth of Li dendrites inside existing dry cracks. Crack propagation depends on the macroscopic fracture toughness of the electrolyte, the current density, stack pressure and maximum length a Li dendrite attains in the crack during a given plating step. Even if the same capacity is plated and stripped, stack pressure results in net lengthening of Li within a dry crack per cycle. Low stack pressures minimize dendrite lengthening and mitigate crack propagation. Cells at stack pressures closer to atmospheric can deliver larger numbers of cycles without Li reaching the counter electrode. A Li/Li_6PS_5Cl cell plated at 4.0 mA cm^{-2} under 7 MPa stack pressure short-circuited after only 35 cycles, whereas a cell at the same current could sustain a fivefold increase in cycling, not short-circuiting until the 170th cycle under atmospheric pressure. This demonstrates the detrimental effect of pressurizing Li anodes during charging.

In practical cells, pressures can result from sources other than stack pressure, for example, intercalation cathodes may produce substantial internal stresses during charge and discharge at high rates. Our model of dendrites not only separates initiation from propagation but the propagation part of the model also explains how, if stresses in the cell induce cracking of the electrolyte and lithium metal finds its way into such pre-existing cracks, the cracks would propagate and induce failure. Low pressures can result in void formation on stripping[4]. Nevertheless, the basic mechanism of initiation and propagation on plating are expected to be similar; the voiding will reduce the contact area and increase the local currents. The desirability of low pressures to suppress dendrite propagation may be incompatible with mitigating voiding, although of course practical cells will have to operate at low stack pressures in any case. Ultimately, any practical solid electrolyte will be tens of microns in thickness. Although the electrolytes used in this study are approximately 1 mm, we anticipate that similar basic processes will occur in thinner electrolytes, even though the dimensions of the features involved (cracks, pores, grain size) will be smaller. Although smaller features may reduce the J-integral, other factors such as more constrained flow of Li metal within smaller features may increase it. Which wins would depend on the details of the materials and operating conditions.

An interesting recent paper has suggested that small-aspect-ratio Li anodes may exhibit different mechanical properties, including creep and flow[30]. This might depend on the adhesion of Li to the specific solid

electrolyte, in that case LLZO. We have not seen evidence of this in our results, which may be because of a different solid electrolyte or conditions used. Different solid electrolytes will differ in their microstructure (grain size, pore size, pore population density) and mechanical properties (fracture toughness and local fracture strength), the latter especially when comparing oxides with sulfides or halides. As a result, the values of these parameters will vary; however, we anticipate the fundamental mechanism of initiation and propagation described here to apply. Li_6PS_5Cl forms a self-limiting solid electrolyte interphase of <250 nm in contact with Li metal[31]. The pores in which dendrites initiate are deeper within the solid electrolyte (pores very near the surface would not build sufficient pressure to initiate dendrites). Therefore, the main effect of a solid electrolyte interphase is probably confined to the density of microcracks connecting the anode with the subsurface pores. In the case of current density being sufficiently low that Li fills the entire length of the crack before reaching the critical length for crack propagation, the mode of cracking will probably switch to Li-tip-driven mechanisms of the type described previously[5,9]. Therefore, this study offers a unified view of dendrites in ceramic electrolytes. Future studies should investigate the consequences of the results reported here, including increasing local fracture strength at the grain boundaries and whether low pressures at the Li anode can be maintained in practical cells, leading to higher critical currents and extended cycling.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-023-05970-4.

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Methods

Electrolyte preparation and cell assembly

All procedures were performed inside an Ar-filled glovebox (O_2 and H_2O levels < 0.1 ppm). Li_6PS_5Cl solid electrolyte powder with a particle size of D50 ≈ 10 μm was purchased from MSE Supplies and pressed into a disc with a 5-mm-diameter stainless-steel die set under a uniaxial pressure of 400 MPa. The disc was then transferred to a furnace and sintered at 300 °C for 20 min. The sintered electrolyte disc was then cooled to room temperature and used for cell assembly. For operando XCT characterization, metallic lithium 100 μm thick was punched into 2-mm-diameter disc-shaped lithium electrodes and pressed onto both sides of the sintered Li_6PS_5Cl disc. The cell was then assembled into a customized tube cell, which maintained airtightness and a stack pressure of 7 MPa on the beamline.

Three-electrode cell cycling

For a three-electrode cell, two disc-shaped lithium electrodes of 1 mm diameter were pressed onto the same side of the electrolyte disc as the working and reference electrodes. The working and reference electrodes were separated by a 1-mm-wide polymer insulating tape mask. A 5-mm lithium disc was pressed onto the other side as the counter electrode. The assembly was then placed into a pouch cell and sealed under vacuum (approximately 10^{-3} bar). Poor Li–solid electrolyte contact can lead to increased critical current densities with increasing pressure because the contact improves with pressure. Therefore, establishing and retaining good contact at the Li–solid electrolyte interface is important when investigating the fundamental mechanism of dendritic failure[27–29]. Therefore, a high forming pressure was used to ensure a good Li/Li_6PS_5Cl contact (see FIB-SEM; Supplementary Fig. 18). The cells were cycled galvanostatically with a capacity on each half-cycle of 0.5 mAh cm^{-2}. Although the plating current on the working electrode was 4.0 mA cm^{-2}, a current density of 0.05 mA cm^{-2} was used on stripping to ensure no void formation at the working electrode–solid electrolyte interface. All galvanostatic measurements were performed using a Gamry Interface 1010 potentiostat.

Operando XCT

X-ray tomograms were recorded at the I13-2 beamline at the Diamond Light Source. A pink beam with beam energy ranging from 8 keV to 30 keV was used to maximize the X-ray beam flux and so improve the time resolution. Projections were collected with a PCO pco.edge 5.5 sCMOS camera combined with an optical microscope with a 4× magnification, resulting in a pixel size of 1.625 μm and a field of view of 4.7 × 4.4 mm^2. A 500-μm-thick LuAG scintillator was used to convert X-rays to visible light. The exposure time was set to 100 ms. For each tomogram, 1,801 equiangularly distributed projections were taken over 180°, as well as 50 dark and 50 flat references before and after the whole operando experiment. X-ray tomograms were obtained at the pristine state and then collected continuously during galvanostatic plating with a current density of 3.0 mA cm^{-2} and a capacity of 3.0 mAh cm^{-2}. During the galvanostatic plating, 7 MPa of pressure was applied using a constantly applied spring force, which was measured and calibrated using a transducer. A final tomogram was collected at the end of plating.

On-line mass spectrometry

On-line mass spectrometry measurements were conducted on pristine and Li-plated Li_6PS_5Cl. Sintered Li_6PS_5Cl discs were placed between Li electrodes and Li plated galvanostatically at a current density of 0.05 mA cm^{-2} and a capacity of 2.0 mAh cm^{-2}. A second cell was assembled identically and no current was passed. The cells were then disassembled and the lithium electrodes removed with 4,000-grit polishing papers in an Ar-filled glovebox. The solid-electrolyte discs were then transferred into a sealed vial along with a magnetic stirrer. An excess amount of degassed aqueous LiOH solution was then injected into the vial, slowly dissolving the Li_6PS_5Cl electrolyte and reacting with any Li metal inside the electrolyte evolving H_2. The released H_2 was detected and quantified by a mass spectrometer (Prima BT, Thermo Fisher Scientific) as a percentage of the total gas flow.

Plasma FIB-SEM

A 5-mm pristine Li_6PS_5Cl disc was transferred into the chamber of a plasma FIB scanning electron microscope (Helios G4 PFIB DualBeam, Thermo Fisher Scientific) for FIB milling with an ILoad vacuum sample transfer system without exposure to air. The freshly milled cross-section was then imaged with SEM to visualize the pores and microcracks in proximity to the interface.

SIMS

A 5-mm Li_6PS_5Cl disc that had undergone a single plating at 0.05 mA cm^{-2} for 5.0 mAh cm^{-2} was imaged using the plasma FIB-SEM procedure above. We performed SIMS analysis on the cross-section using a Hiden Analytical EQS electrostatic quadrupole SIMS detector. A focussed beam of Xe^+ was used for SIMS analysis. The quadrupole detector was tuned to collect the $^7Li^+$ signal.

Nanoindentation: elastic modulus, hardness and fracture toughness

In situ nanoindentation and microcantilever bending (Hysitron PI 88) in a TESCAN scanning electron microscope enabled measurement of air-sensitive Li_6PS_5Cl discs without the need for mineral oil or other techniques that can affect the mechanical response[32]. Samples were transferred to the scanning electron microscope in an airtight transfer container. Indentation modulus and hardness were calculated using the Oliver–Pharr method[33] on several indents made with a Berkovich tip. Further details can be found in the Supplementary Information.

For the determination of fracture toughness, a cube corner tip was used to cause fracture of the electrolyte. Using the Lawn, Evans and Marshall equation (equation (1)), the fracture toughness, K_{IC}, was calculated for each of the indents. In this equation, E is modulus, H is hardness, P is load, c is total crack length and α is a geometric constant for a cube corner tip. The average across all indents was taken as the Li_6PS_5Cl fracture toughness.

$$K_{IC} = \alpha\left(\frac{E}{H}\right)^{1/2}\left(\frac{P}{c^{3/2}}\right) \quad (1)$$

The standard protocols for determining fracture toughness were followed, based on refs. 34,35, including ensuring minimal pileup, as shown in Supplementary Fig. 21. Further details are available in the Supplementary Information.

Microcantilever bending: local fracture strength

Microcantilevers were produced using a Helios G4 PFIB DualBeam system (Thermo Fisher Scientific) combined with an ILoad vacuum sample transfer system. The preparation of microcantilevers using the FIB milling method is described in detail in previous work[36]. Loading was performed using a cube corner tip, in a displacement-controlled regime, at a rate of 2.5 nm s^{-1} until fracture occurred. The grain boundary strength σ_c for each microcantilever was calculated using the following equation for the tensile surface stress:

$$\sigma_c = \frac{P_f L_C y}{I}$$

in which P_f is load at the point of fracture, L_C is the distance between the loading point and the crack, I is the second moment of area and y is the vertical distance between the neutral plane and the upper surface. I and y were calculated for a pentagonal beam as shown by Chen et al.[37]. Three microcantilevers were tested and the mean value of the tensile surface stress at failure was taken for the local fracture strength of the Li_6PS_5Cl.

Data availability

The datasets generated and/or analysed during this study are available from the corresponding author on reasonable request.

Code availability

The computer code generated and used during this study is available from the corresponding author on reasonable request.

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Author contributions Z.N., G.L. and D.L.R.M. contributed to all aspects of the research. Z.N., D.L.R.M., D.S.-J., S.D.P., G.O.H. and A.J.B. carried out the operando synchrotron XCT. Z.N. and D.L.R.M. performed the preparation of electrolyte discs and cell assembly. Z.N., D.L.R.M, C.G. and X.G. performed the on-line mass spectrometry. Z.N., D.L.R.M., B.H., B.L. and J.B. performed the plasma FIB imaging. D.L.R.M. and J.B. performed plasma FIB imaging with SIMS. Z.N., D.L.R.M., J.P., J.L. and D.E.J.A. conducted the preparation of microcantilever and mechanical tests. G.L., Y.C. and C.W.M. conducted the modelling. Z.N., G.L., D.L.R.M., D.S.-J., R.I.T., P.S.G., D.E.J.A., T.J.M., C.W.M. and P.G.B. discussed the data. All authors contributed to the interpretation of data. Z.N., D.L.R.M., G.L., C.W.M. and P.G.B. wrote the manuscript, with contributions and revisions from all authors. The project was supervised by C.W.M., T.J.M. and P.G.B.

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Additional information

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